# COMMUTING COUNTEREXAMPLES TO DETERMINANTAL ENTROPY CONCAVITY

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ABSTRACT. This note records a statement refuted in the *Count ex Machina* archive: each is quoted as posed and answered by an explicit witness, and the archive carries a certificate that recomputes the refutation by rigorous interval arithmetic.

*Summary:* The exact-set Shannon entropy of a determinantal law is midpoint concave in its positive-contraction kernel. *Certificate:* Commuting $5 \times 5$ Hermitian kernels with rational spectra; rigorous rational logarithm bounds certify an entropy midpoint deficit exceeding 1/5,000,000,000.

*Posed by Russell Lyons (2003; 2014); counterexample found by GPT Astra (Pro) (2026-10-30); formalized by OpenAI Codex; audited by OpenAI Codex; contributed by Suvrit Sra.*

Fix a finite coordinate set $E$. A positive contraction $Q$ is a self-adjoint operator on the real or complex Hilbert space $\ell^2(E)$ satisfying $0 \preceq Q \preceq I$. Its determinantal law $\mathbf{P}^Q$ is defined by $\mathbf{P}^Q[A \subseteq X] = \det Q_{A,A}$ for every $A \subseteq E$. The entropy in the statement is the Shannon entropy of the entire occupied set, with the coordinate basis fixed:

$$\mathrm{Ent}(Q) = -\sum_{S \subseteq E} \mathbf{P}^Q(\{S\}) \log \mathbf{P}^Q(\{S\}), \qquad 0 \log 0 = 0.$$

We use natural logarithms. Write $H(Q) = \mathrm{Ent}(Q)$ below. For the coordinate projection $P_{S^c}$, the exact-set convention is

$$p_S(Q) := \mathbf{P}^Q(\{S\}) = (-1)^{|S^c|} \det(Q - P_{S^c}).$$

This is not the entropy of the particle count and not the sum of binary entropies of the eigenvalues. The source explicitly permits complex Hilbert spaces (Section 2.1 of (Lyons, 2014)), so complex Hermitian kernels are within the stated domain; commutativity is a property of our witness, not a hypothesis of the statement.

**Problem 1** (As posed: R. Lyons (2014), Conjecture 2.6, eq. (2.14); first posed as Conjecture 9.2 (Lyons, 2003))**.** For all positive contractions $Q_1$ and $Q_2$, we have

$$\mathrm{Ent}\big((Q_1 + Q_2)/2\big) \geq \big(\mathrm{Ent}(Q_1) + \mathrm{Ent}(Q_2)\big)/2.$$

The entropy conjecture of (Lyons, 2014) fails along a commuting affine family. The matrices have an exact Fourier eigenbasis and rational spectra; all squared minors needed for their occupation probabilities lie in $\mathbb{Q}(\sqrt{5})$.

**Theorem 2** (**[Disproof]**: determinantal entropy concavity)**.** *Let* $E = \{0, 1, 2, 3, 4\}$*, let*

$$\omega = e^{2\pi i/5}, \qquad F_{jk} = 5^{-1/2} \omega^{jk} \quad (0 \leq j, k \leq 4),$$

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From the *Count ex Machina* archive of counterexamples, https://github.com/suvrit/count-ex-machina, whose companion paper is arXiv:2608.29595.

*and set*

$$\lambda_+ = \frac{1}{10000}(11, 9, 9991, 9990, 9989), \qquad \lambda_- = \frac{1}{10000}(9, 11, 9989, 9990, 9991),$$

*and* $K_\pm = F\,\mathrm{diag}(\lambda_\pm)F^*$. *These are commuting Hermitian strict positive contractions, but*

$$H(K_+) = H(K_-), \qquad H(K_+) - H\left(\frac{K_+ + K_-}{2}\right) > \frac{1}{5{,}000{,}000{,}000} > 0.$$

*In particular Conjecture 2.6 of (Lyons, 2014) is false as stated.*

*Proof.* All displayed spectral entries are exact rational numbers. Because $F$ is unitary, the two matrices commute, and their eigenvalues are strictly between zero and one. Introduce the family

$$\lambda_\varepsilon(t) = (\varepsilon + t, \varepsilon - t, 1 - \varepsilon + t, 1 - \varepsilon, 1 - \varepsilon - t), \qquad K_\varepsilon(t) = F\,\mathrm{diag}(\lambda_\varepsilon(t))F^*.$$

The theorem uses $\varepsilon = 1/1000$ and $t = \pm 1/10000$. For every $0 < \varepsilon < 1/2$ and $|t| < \varepsilon$, the entire family consists of commuting strict positive contractions.

For $J \subseteq E$ put

$$q_J(\lambda) = \prod_{j \in J} \lambda_j \prod_{j \notin J} (1 - \lambda_j), \qquad M(S, J) = |\det F_{S,J}|^2.$$

The exact-set probabilities satisfy

$$p_S(F\,\mathrm{diag}(\lambda)F^*) = \sum_{J:\,|J|=|S|} M(S, J) q_J(\lambda). \tag{1}$$

Indeed the inclusion convention gives the probability generating polynomial $\det(I - K + KZ)$, with $Z = \mathrm{diag}(z_i)$. Conjugating by $F$ and expanding in the rows of $I - \mathrm{diag}(\lambda)$ and $\mathrm{diag}(\lambda)F^*ZF$ gives $\sum_J q_J(\lambda) \det((F^*ZF)_{J,J})$. Cauchy–Binet expands the latter minor as $\sum_{|S|=|J|} M(S, J) \prod_{i \in S} z_i$; comparison of coefficients proves (1).

Write $s = \sqrt{5}$. Squared Fourier minors of sizes zero and five equal one; those of sizes one and four equal $1/5$. For two-element sets $S = \{a, b\}$ and $J = \{c, d\}$,

$$M(S, J) = \frac{|\omega^{ac+bd} - \omega^{ad+bc}|^2}{25} = \begin{cases} (5 - s)/50, & (b - a)(d - c) \equiv \pm 1 \pmod 5, \\ (5 + s)/50, & (b - a)(d - c) \equiv \pm 2 \pmod 5. \end{cases}$$

The size-three values are the complementary size-two values, by the complementary-minor identity for a unitary matrix. Thus every $p_S$ is an explicit polynomial in $\varepsilon, t$ with coefficients in $\mathbb{Q}(\sqrt{5})$.

Let $R$ permute eigenmodes by $j \mapsto 1 - j \pmod 5$. It fixes $\lambda_\varepsilon(0)$ and reverses the direction $(1, -1, 1, 0, -1)$. Since $FR = D\overline{F}$, where $D_{jj} = \omega^j$, we have

$$K_\varepsilon(-t) = D\overline{K_\varepsilon(t)}D^*.$$

Complex conjugation and diagonal unitary conjugation preserve each exact-set probability. Hence every $p_S(\varepsilon, t)$ is even in $t$, proving the equality of the endpoint distributions and entropies.

It remains to establish the strictly positive finite entropy difference. The accompanying `verify.py` evaluates (1) with exact rational coefficients and encloses all logarithms by rational bounds. It computes the eigenvalues at both endpoints and checks strict admissibility, polynomial normalization, and coefficientwise evenness. Its enclosure $[L, U]$ for $H(K_+) - H((K_+ + K_-)/2)$ satisfies the exact assertions

$$L > \frac{1}{5{,}000{,}000{,}000} > 0, \qquad L \le H(K_+) - H\left(\frac{K_+ + K_-}{2}\right) \le U.$$

Here is the complete error-control method used by that finite computation. With $D_0 = 10^{40}$ and $r = \lfloor\sqrt{5D_0^2}\rfloor$, integer square root gives

$$r/D_0 < \sqrt{5} < (r + 1)/D_0.$$

After exact power-of-two range reduction, logarithms of positive rational endpoints reduce to arguments $1 \le x \le 2$. For $z = (x-1)/(x+1)$ and $N = 40$,

$$\log x = 2\sum_{k=0}^{N-1} \frac{z^{2k+1}}{2k+1} + R_N, \qquad 0 \le R_N \le \frac{2z^{2N+1}}{(2N+1)(1-z^2)}.$$

The series follows by integrating the geometric series; the tail bound replaces each omitted denominator by $2N+1$. Algebraic arguments are enclosed using the bounds on $\sqrt{5}$, and logarithms by monotonicity. Interval multiplication uses the minimum and maximum of all four endpoint products. Endpoints are rounded outward, using integer floor and ceiling, to a grid of denominator $10^{32}$. These operations preserve containment at each step. Every calculation and decisive inequality uses integers or `fractions.Fraction`; no floating-point evidence is used. This proves the stated finite inequality. □

There is also a complete analytic existence proof for the same family, independent of executing the certificate. It explains the sign of the curvature near a rank-three projection.

**Proposition 3.** *For the family above,*

$$\lim_{\varepsilon\downarrow 0} \frac{d^2}{dt^2} H(K_\varepsilon(t))\bigg|_{t=0} = 2\log\frac{15}{44} + \frac{2}{\sqrt{5}}\log\frac{147+65\sqrt{5}}{22} > 0.$$

*Proof.* The kernel is circulant, so translating $S$ preserves its probability. For the eight translation classes write

$$p_S(\varepsilon, 0) = c_S\varepsilon^{m_S}(1+O(\varepsilon)), \qquad \partial_t^2 p_S(\varepsilon, 0) = d_S + O(\varepsilon).$$

The Fourier-minor formula gives the following table; the last column is per individual outcome, before multiplication by the class size.

| $S$ | class size | $m_S$ | $c_S$ | $d_S$ |
|---|---|---|---|---|
| $\varnothing$ | 1 | 3 | 1 | 0 |
| $\{0\}$ | 5 | 2 | $3/5$ | $-2/5$ |
| $\{0,1\}$ | 5 | 1 | $(15-s)/50$ | $(10-2s)/25$ |
| $\{0,2\}$ | 5 | 1 | $(15+s)/50$ | $(10+2s)/25$ |
| $\{0,1,2\}$ | 5 | 0 | $(5-s)/50$ | $(-10+6s)/25$ |
| $\{0,1,3\}$ | 5 | 0 | $(5+s)/50$ | $(-10-6s)/25$ |
| $\{0,1,2,3\}$ | 5 | 1 | $2/5$ | $4/5$ |
| $E$ | 1 | 2 | 1 | $-2$ |

For completeness, the leading latent set at $\varepsilon = t = 0$ is $J_0 = \{2,3,4\}$. The leading singleton contributions are its three singletons, giving $3/5$. The leading pair contributions are $\{2,3\}, \{2,4\}, \{3,4\}$, whose squared minors sum to $(15 \mp s)/50$. The leading triple contribution is $J_0$ itself. The two leading four-point sets add either 0 or 1 to $J_0$, giving $2/5$. The empty and full probabilities at $t = 0$ are $\varepsilon^3(1-\varepsilon)^2$ and $\varepsilon^2(1-\varepsilon)^3$. These facts verify $m_S, c_S$.

To check $d_S$ directly, the only nonzero values of $\partial_t^2 q_J(\lambda_\varepsilon(t))$ at $\varepsilon = t = 0$ are

| $J$ | $q_J''$ | $J$ | $q_J''$ |
|---|---|---|---|
| $\{3\}$ | $-2$ | $\{2,3\}$ | 2 |
| $\{3,4\}$ | 2 | $\{0,2,3\}$ | 2 |
| $\{1,2,3\}$ | $-2$ | $\{0,3,4\}$ | $-2$ |
| $\{1,3,4\}$ | 2 | $\{2,3,4\}$ | $-4$ |
| $\{0,2,3,4\}$ | 2 | $\{1,2,3,4\}$ | 2 |
| $E$ | $-2$ | | |

These are obtained by differentiating the five-factor products defining $q_J$. Multiplication by the squared Fourier minors and summation gives the displayed $d_S$ values. Polynomial dependence supplies all stated error terms.

At every positive $\varepsilon$, all probabilities are positive, so entropy is smooth. Evenness gives $p'_S(\varepsilon,0)=0$, and normalization gives $\sum_S p''_S=0$. Therefore

$$H''_\varepsilon(0) = -\sum_S p''_S(\varepsilon,0)\log p_S(\varepsilon,0).$$

The potential $\log\varepsilon$ terms cancel, because $\sum_S m_S d_S = -4+4+4-4=0$; all errors are $O(\varepsilon\log\varepsilon)$ or smaller and vanish. Put $a=(5-s)/50$, $b=(5+s)/50$, $c=(15-s)/50$, $d=(15+s)/50$. The finite limit is

$$\begin{aligned} C &= 2\log(3/5) - 4\log(2/5) \\ &\quad - \tfrac{2}{5}\big[(5-s)\log c + (5+s)\log d\big] + \tfrac{2}{5}\big[(5-3s)\log a + (5+3s)\log b\big] \\ &= 2\log(15/44) + \tfrac{2}{s}\log\big((147+65s)/22\big), \end{aligned}$$

using $ab=1/125$, $cd=11/125$, and $(c/d)(b/a)^3=(147+65s)/22$. Since $15/44>1/3$, $(147+65\sqrt{5})/22>13$, and $2/\sqrt{5}>8/9$,

$$C > \tfrac{8}{9}\log 13 - 2\log 3 = \tfrac{2}{9}\log(28561/19683) > 0.$$

The second and third inequalities follow from $11/5<\sqrt{5}<9/4$. Thus positive curvature occurs at interior kernels for every sufficiently small positive $\varepsilon$. No boundary derivative of entropy is used. □

*Remark* 4 (Scope). The finite witness refutes the source's midpoint inequality itself, with its original hypotheses and quantifiers intact. It also refutes full concavity, even on a commuting affine family. It does not establish the least possible dimension or settle the real-symmetric restriction. It concerns finite exact-set Shannon entropy; it does not refute concavity of particle-count entropy, spectral entropy, or any statement about entropy rates of infinite stationary processes.

As a consequence, the auxiliary balanced beam-splitter inequality from the discovery discussion also fails for this pair. For

$$W = \frac{1}{\sqrt{2}}\begin{pmatrix} I & I \\ -I & I \end{pmatrix}, \qquad Q = W(K_+ \oplus K_-)W^*, \qquad \bar{K} = (K_+ + K_-)/2,$$

both output coordinate marginals have kernel $\bar{K}$. Subadditivity yields $H(Q) \le 2H(\bar{K}) < H(K_+) + H(K_-)$, with output-minus-input gap strictly less than $-1/2{,}500{,}000{,}000$. This is a corollary, not a second provenance claim or an attribution of that auxiliary statement to Lyons.

## Code and certificate

The case is the directory `counterexamples/lyons-dpp-entropy-concavity/` of the archive, at commit `73b416e`. Its `verify.py` recomputes every refutation in this note using exact arithmetic or rigorous interval enclosures, never floating-point evidence alone; it raises an error if any check fails and otherwise writes the certificate below. To rerun it:

```
git clone https://github.com/suvrit/count-ex-machina
cd count-ex-machina/counterexamples/lyons-dpp-entropy-concavity
python verify.py
```

- Theorem 2 (result `lyons-dpp-entropy-concavity`): rigorous interval arithmetic.
- `artifacts/certificate.json`: Exact spectra, occupation probability coefficients, and outward rational intervals certifying the entropy midpoint violation and positive curvature.

## AI USAGE

The credit line under the title records, for each statement, who posed it, who found its counterexample and when, and who formalized and audited it; the counterexample here was found by GPT Astra (Pro) (2026-10-30). This note was assembled mechanically from the case's source in the archive by `tools/notes.py`, written with Claude Code; its mathematics is exactly that of the archive at the commit cited above.

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## REFERENCES

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